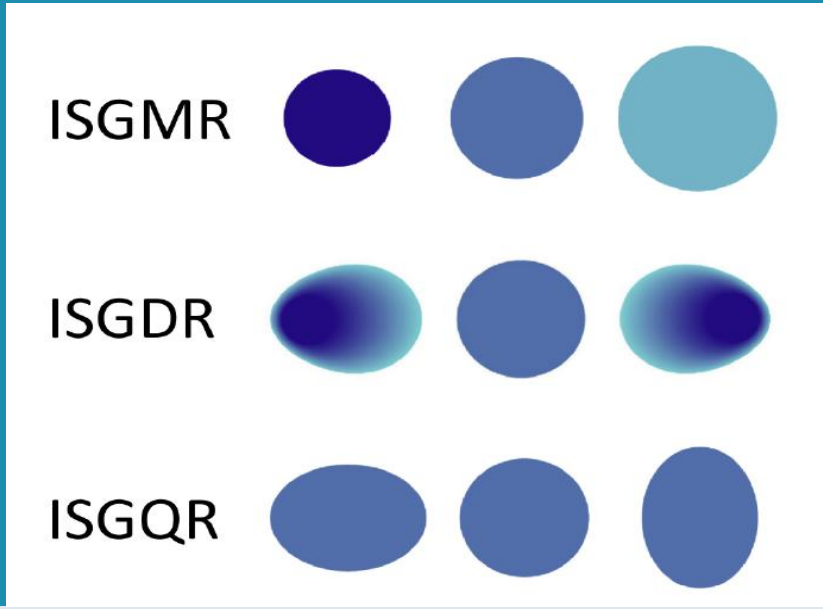
KU LEUVEN

Measurement of isoscalar resonances in 58Ni



Alex Antony Arokiaraj, 27 June 2025

Supervised by Prof. Dr. Riccardo Raabe, Dr. Marine Vandenbrouck

Giant Resonances

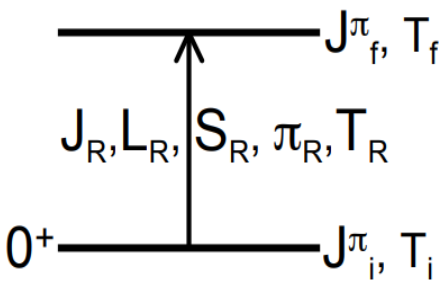
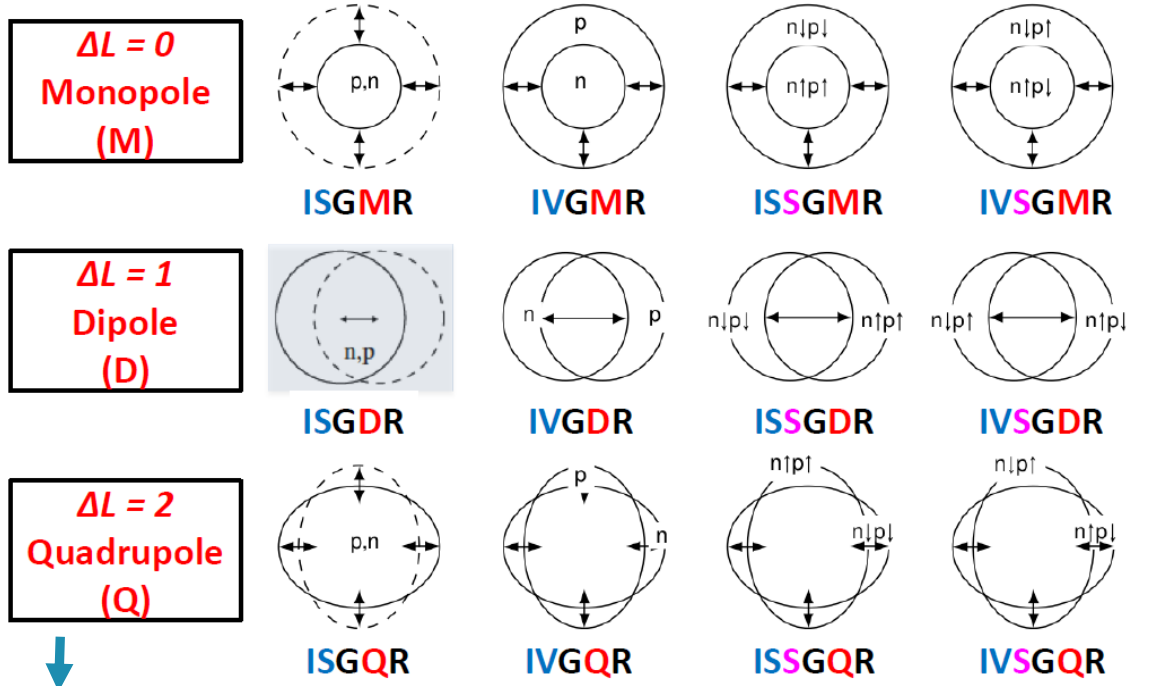


Fig courtesy: Elias Khan



Multipole Decomposition Analysis

Inelastic scattering – alpha probe

ΔT = 0 Isoscalar (IS) ΔS = 0 Electric	ΔT = 1 Isovector (IV) ΔS = 0 Electric	ΔT = 0 Isoscalar (IS) ΔS = 1 Magnetic (S)	ΔT = 1 Isovector (IV) ΔS = 1 Magnetic (S)

Fig courtesy: Bagchi 2015 PhD Thesis

Impetus for GR measurements

$$E_{ISGMR} = \hbar\sqrt{\frac{K_A}{m < r^2 >}}$$

Macroscopic Approach

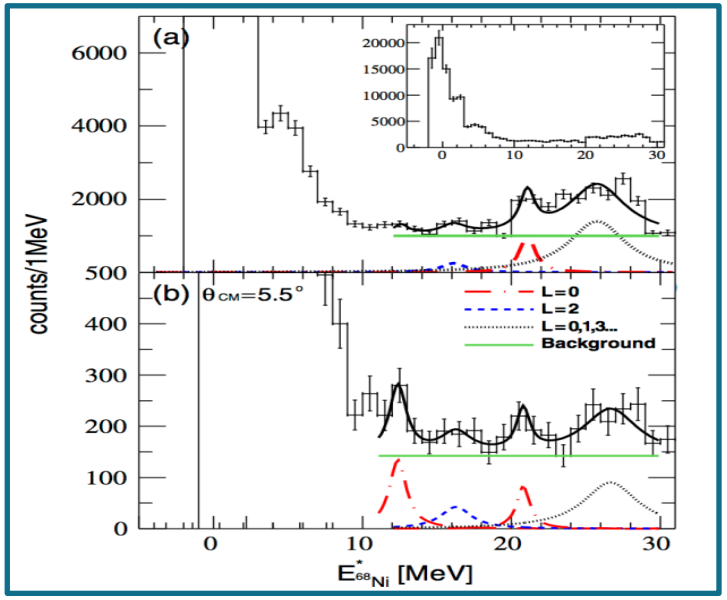
K_∞ = 100-400 MeV *J.P. Blaziot et.al 1995*

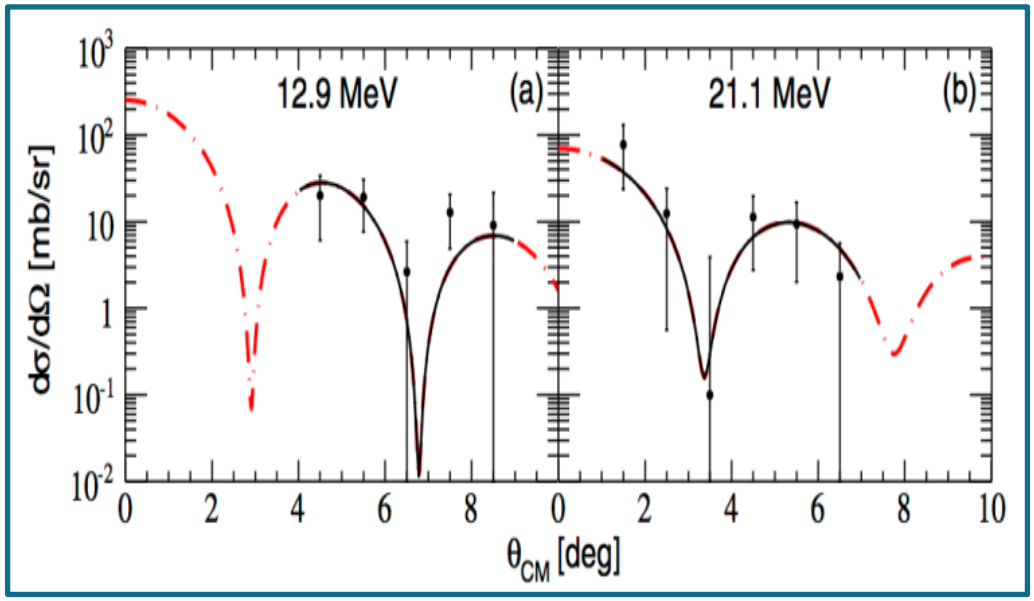
$$K_A = K_\infty + K_{surf} A^{-1/3} + K_\tau \left(\frac{N-Z}{A}\right)^2 + K_{Coul} Z^2 A^{-4/3}$$

Microscopic Approach

- Fully Consistent RPA (Random Phase Approximations)
- Provides at the same time K_∞ , E_{ISGMR}
- K_∞ = 240 ± 10 MeV, ^{208}Pb *B.K.Agrawal et.al 2003*
- K_∞ = 230 ± 40 MeV, ^{208}Pb, ^{120}Sn *Khan et.al 2013*

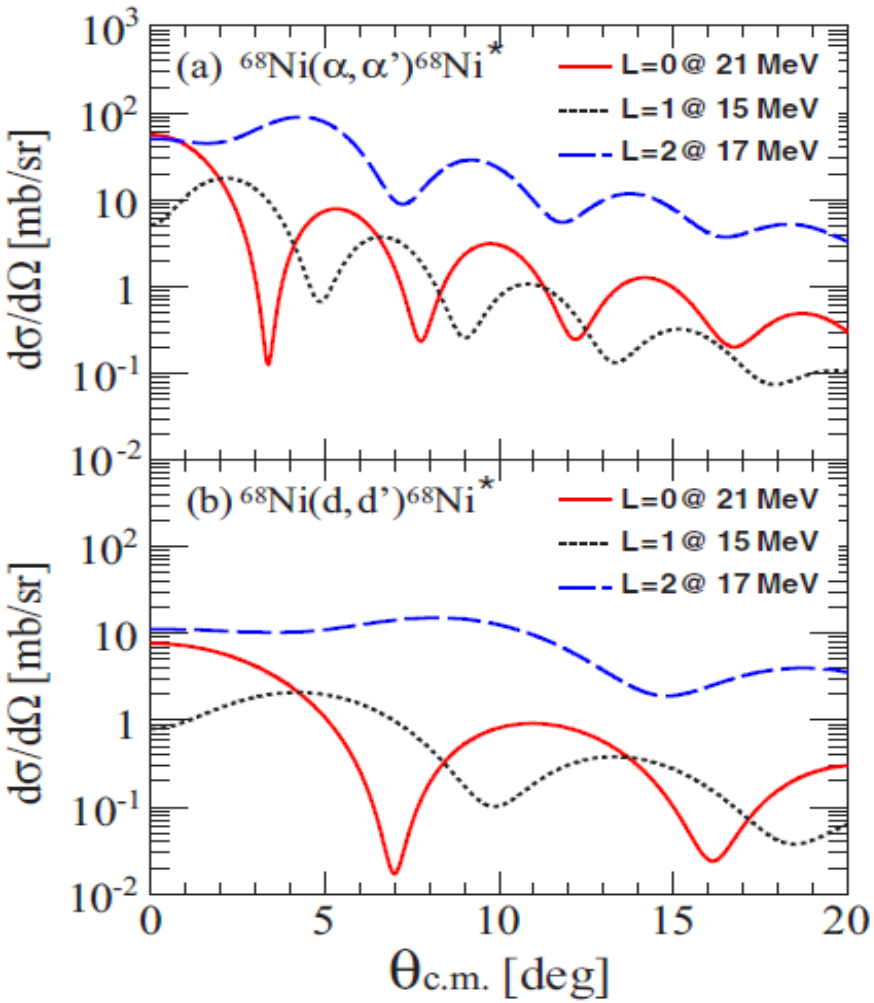
Benchmarks

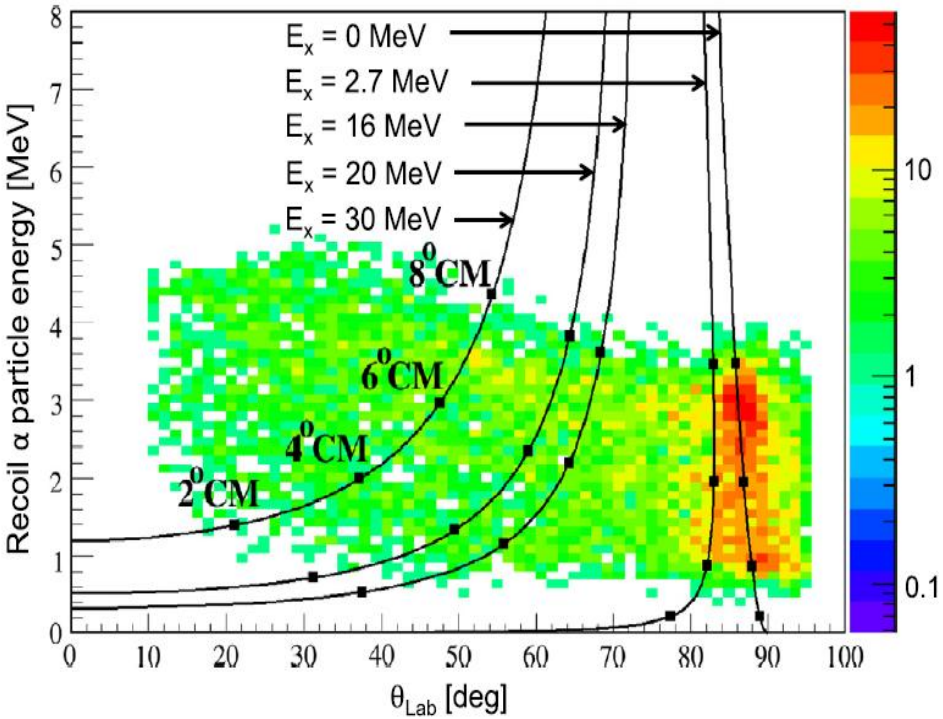




Element	Beam Energy	Work	Measurements	Target
^{56}Ni	50 MeV/A	*Monrozeau et.al 2008*	E_{ISGMR} = 19.3 ± 0.5 MeV E_{ISGQR} = 16.2 ± 0.5 MeV	MAYA
^{68}Ni	50 MeV/A	*Vandebrouck et.al 2015*	E_{ISGMR} = 21.1 ± 1.9 MeV E_{ISGQR} = 15.9 ± 1.3 MeV	MAYA
^{56}Ni	50 MeV/A	*Bagchi et.al 2015*	E_{ISGMR} = 19.1 ± 0.5 MeV E_{ISGDR} = 17.4 ± 0.7 MeV	MAYA

Measurements with MAYA Active Target

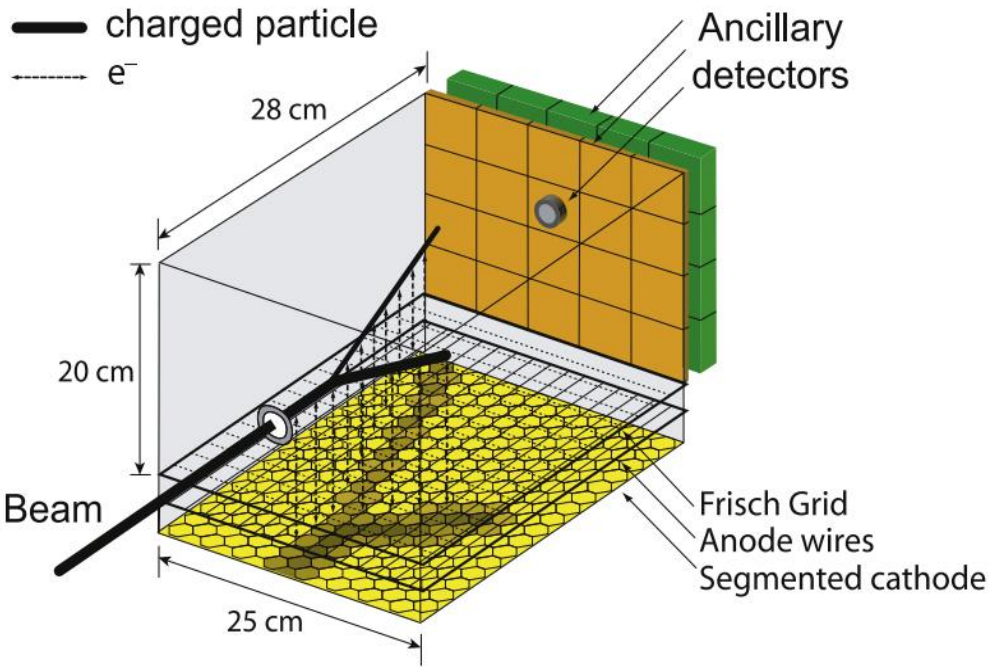


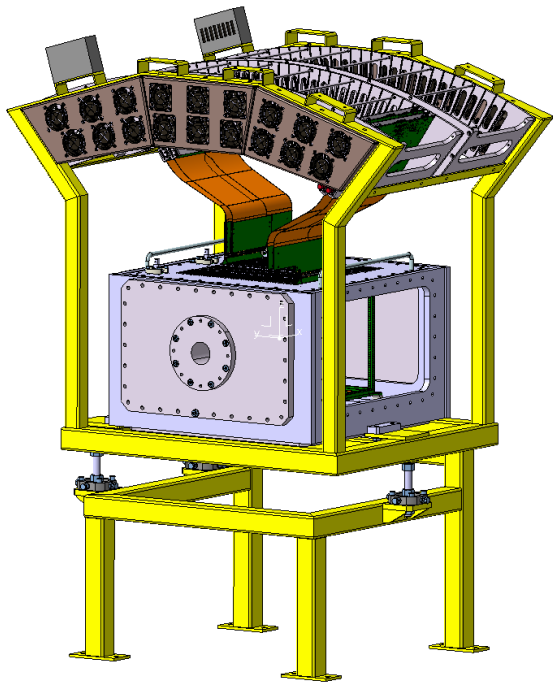


Vandebrouck et.al 2015

Fig courtesy: Bagchi 2015 PhD Thesis

ACTAR TPC – Improved version of MAYA

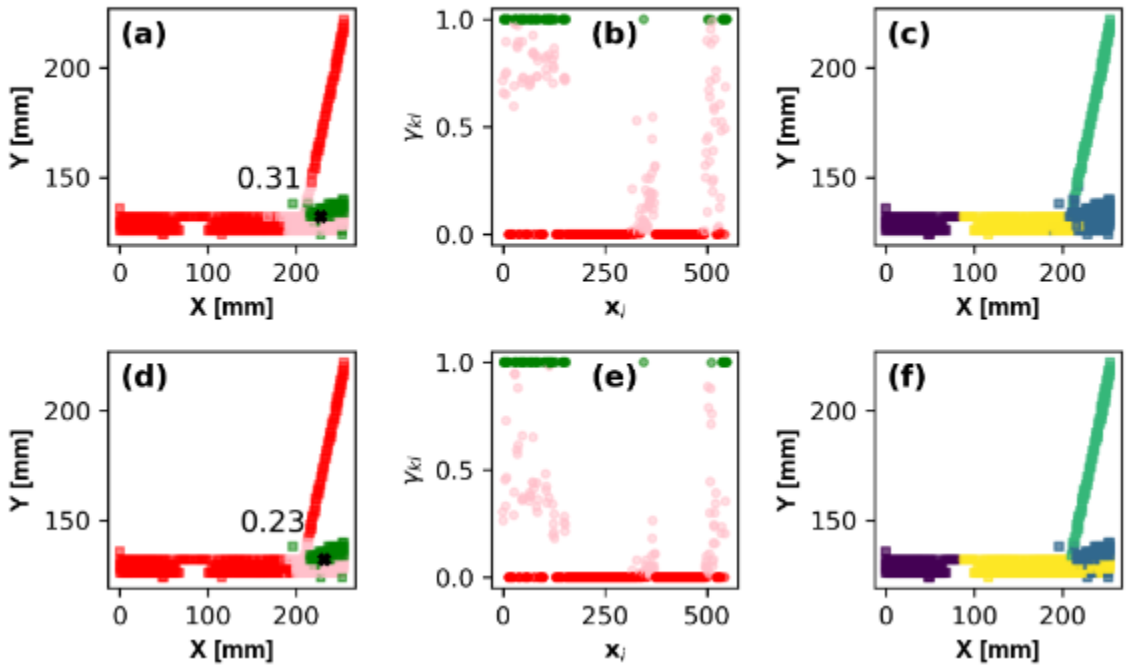




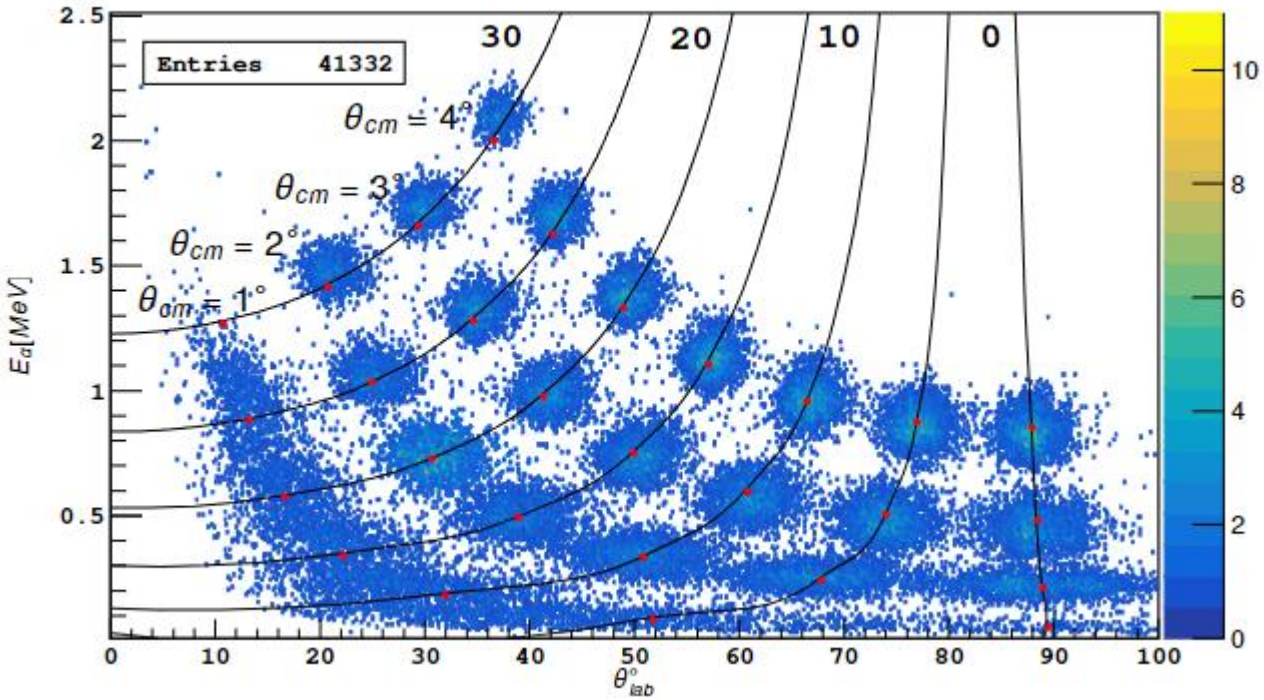
MAYA - 35 × 35 hexagonal pads with 5 mm sides

ACTAR - 128 × 128 square pads with 2 mm sides

Track Reconstruction Techniques determination of lab angle and range

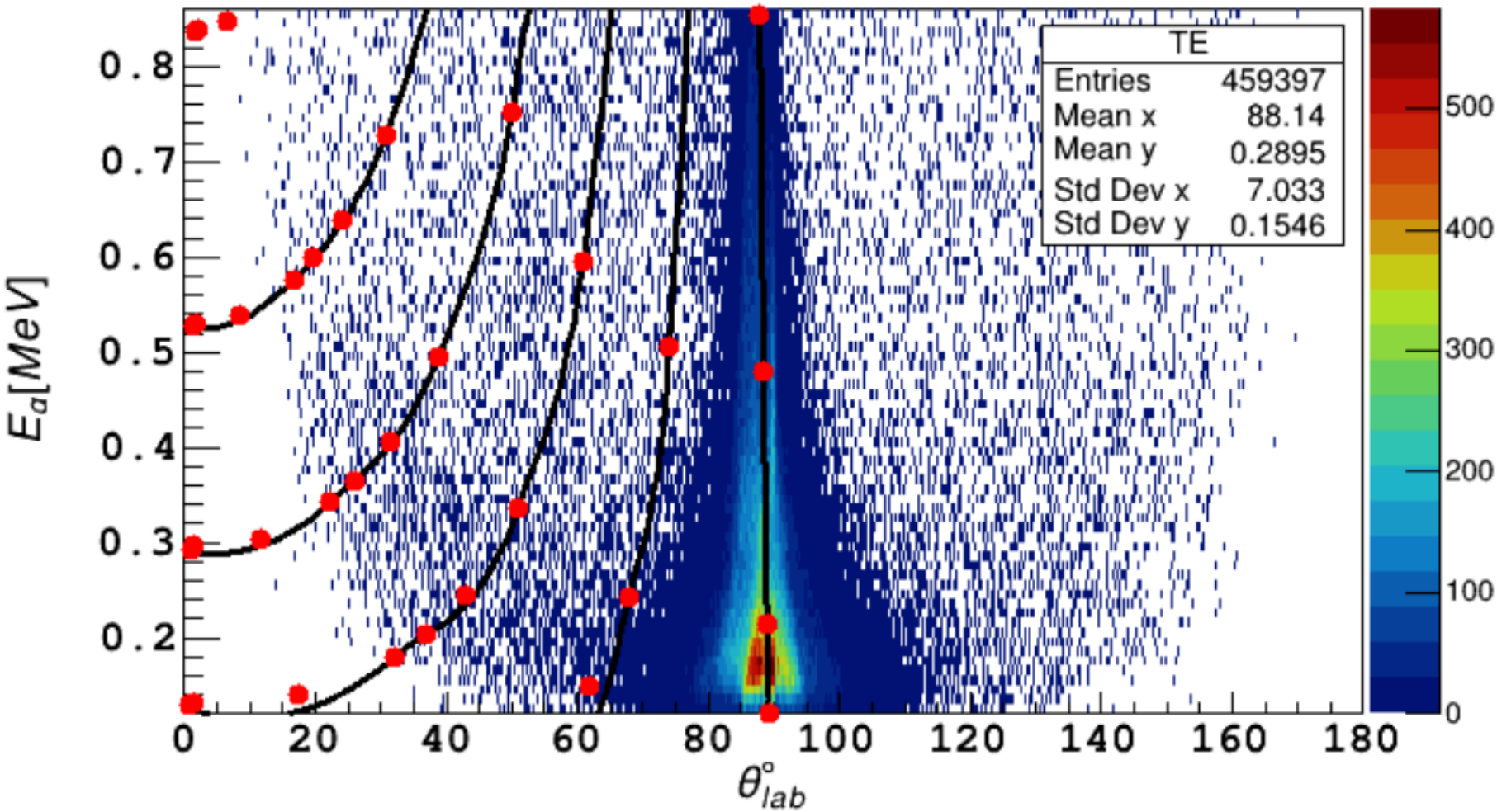


Performance on Simulated Data

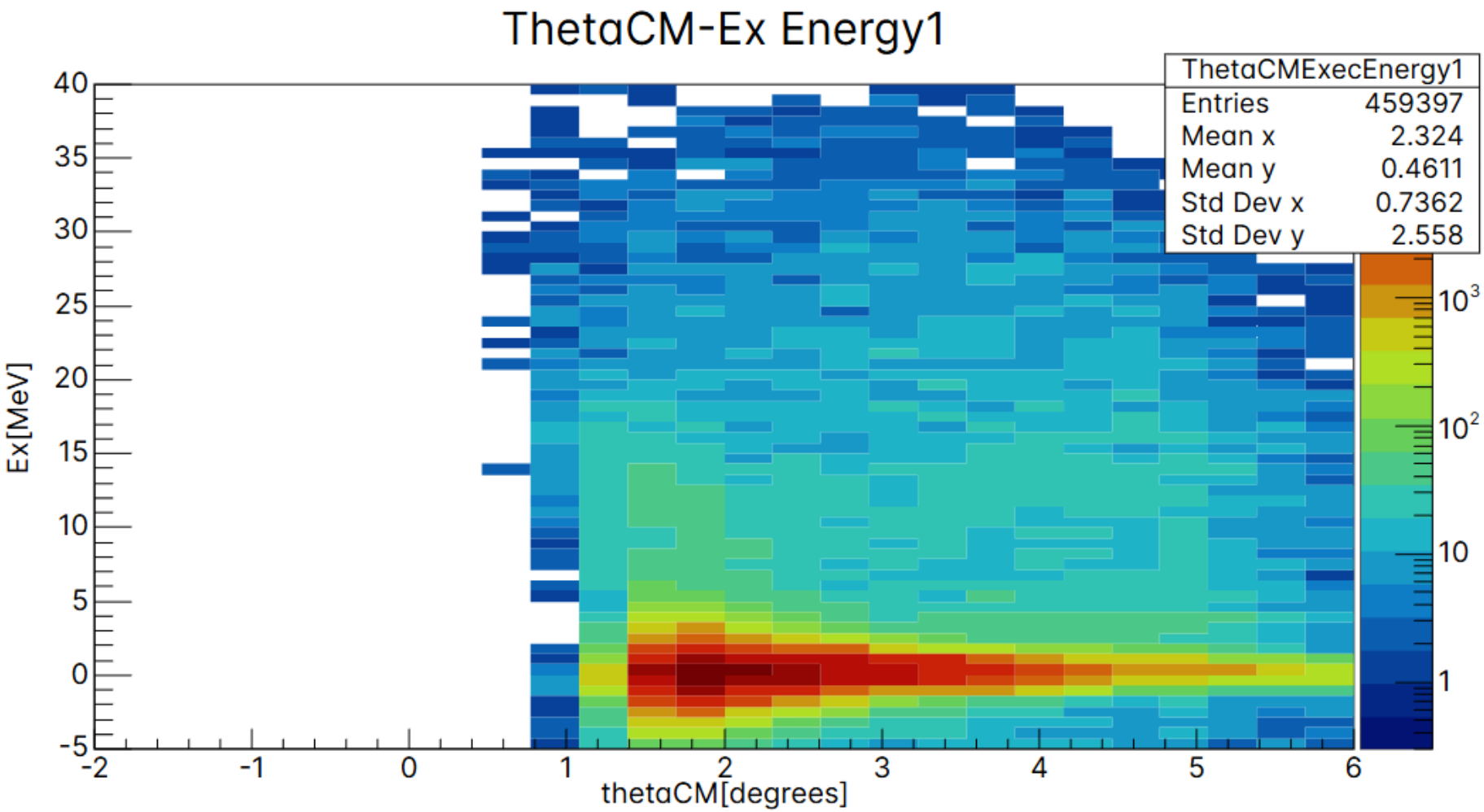


Events reconstructed from the simulation on the kinematic line. The numbers marked on each kinematic line represent the excitation energy in MeV. The filled red circles correspond to the center of mass angles (θcm=1.0◦, 2.0◦, 3.0◦, 4.0◦) for each excitation energy

Validation on Experimental Data



Results on the Experimental Data



Article on the Track Reconstruction

Menu | track_reconstruction_techniques_58ni | Review | Share | Submit | History | Layout | Chat

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- elsarticle.dtx
- elsarticle.ins
- references.bib

File outline

- Introduction
- Generating Simulated Dataset
- Track Reconstruction with GMM
  - Identification of Clusters
  - Regularization
  - Evaluation
  - Reaction vertex and track length
- Case study with simulated \Niaa\ data
  - Event Selection
  - Determination of q
  - Determination of β
  - Validation for the physics case
- Detection Efficiency
  - Evaluation
  - Dealing with bin overflows
- Case study with experimental \Niaa\ data
  - Experimental setup
  - Data Analysis
- Conclusions


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Track reconstruction techniques for analyzing inelastic scattering data acquired using active targets

A. Arokiaraj[a,*], M. Latif[a,1], R. Raabe[a], D. Thisse[b], M. Vandebrouck[b]

[a] KU Leuven, Institute for Nuclear and Radiation Physics, 3001 Leuven, Belgium
[b] CEA Saclay, Saclay, France

Abstract

Active targets like ACTAR TPC [1] are detectors suited for studying giant resonances in unstable nuclei using inelastic scattering in inverse kinematics. One of the major challenges associated with such measurements is the detection of the low-energy ejectiles at low angles with respect to the beam. Detecting such tracks is particularly useful to disentangle different resonance modes. Probabilistic models such as the Gaussian Mixture Model (GMM) are well suited for capturing multiple covariance structures such as the ones existing at the beam-ejectile interface of short-angled tracks. In the present paper, we present track reconstruction techniques built on the GMM for reconstructing the kinematics of inelastic scattering measurements carried out with active targets. The case of low energy tracks are discussed in detail. For the sake of illustration, the techniques are applied to simulated data of ^{58}Ni$(\alpha,\alpha')^{58}$Ni at an incident energy E = 49 MeV/nucleon, and on experimental data for the same reaction acquired using ACTAR TPC.

Keywords: active targets, mixture models, ACTAR TPC, giant resonances, track reconstruction

1. Introduction

The experimental study of giant resonances using active targets has gained attention over the past two decades. The excitation of the isoscalar giant monopole resonance is one of the objectives of such measurements owing to the correlation between its properties and the nuclear matter incompressibility K_∞ [2, 3]. Such measurements are gradually extending to nuclei far from stability. The results are used to understand the effects of neutron-proton asymmetry on the compressibility parameter and to constrain the error bars on the incompressibility of infinite nuclear matter [4]. Alongside such studies, interesting predictions have been made on these nuclei such as the appearance of the soft monopole resonances in the nickel isotopic chain [5, 6]. Only a few setups exist world wide to carry out such studies, in the form of storage rings [7] and time projection chambers [8, 9, 10, 11].

*Corresponding author
[1] Present address:

Preprint submitted to Nuclear Instruments and Methods in Physics Research B *November 20, 2024*

PhD Activities

- Simulations for a paper on the study of the beta delayed alpha particles from 11 Be.
- Master project for a 3 ECTS course (Research Methods in Nuclear Physics)
- Teaching assistant for the Classical Mechanics course
- **Additional course units followed**:
- Managing my PhD
- Data Analysis with Python
- Intensive Academic Writing
- Writing successful grants
- Creative Problem Solving
- Interpersonal Skills
- Introduction to Leadership

Time plan

Expected Timelines

1) Submission of article – **October 2025**
2) Acceptance - **December 2025**
3) Thesis Submission – **March 2026**
4) Completion – **April – May 2026**